

Histologic Chorioamnionitis Is Associated With Reduced Risk of Late-Onset Sepsis in Preterm Infants



WHAT'S KNOWN ON THIS SUBJECT: Chorioamnionitis is associated with preterm birth and an increased risk of common adverse outcomes of prematurity, including early-onset sepsis and neurodevelopmental impairment. The effect of chorioamnionitis on postnatal immune function and risk of late-onset infection is unknown.



WHAT THIS STUDY ADDS: Chorioamnionitis, which complicates many preterm births, is independently associated with a significantly reduced risk of late-onset sepsis in preterm infants. The findings suggest that chorioamnionitis may modulate the development of postnatal immunity in a clinically significant manner.

abstract



BACKGROUND: Histologic chorioamnionitis (HCA) is implicated in the onset of preterm labor and delivery. Chorioamnionitis is a known risk factor for early-onset sepsis and may modulate postnatal immunity. Preterm infants are at greatly increased risk of late-onset sepsis (LOS), particularly with coagulase-negative staphylococci (CoNS), but the impact of HCA on the risk of LOS is unknown.

METHODS: Eight hundred thirty-eight preterm infants born at <30 weeks gestational age at a single tertiary center were included. Histologic examination of placenta and extraplacental membranes was performed, and clinical data were extracted from hospital databases. The influence of HCA on the incidence of early-onset sepsis and LOS was examined using logistic regression analysis and Cox proportional hazards regression.

RESULTS: Mean gestational age was 26.9 ± 1.9 weeks, and mean birth weight was 936 ± 277 g. Two hundred and seventy-six (33%) of 838 infants developed LOS. The presence of fetal or maternal HCA, or maternal HCA and fetal HCA alone, was associated with a significantly decreased risk of LOS with any organism. Histologic chorioamnionitis correlated with a significantly decreased risk of CoNS LOS.

CONCLUSIONS: HCA is associated with a significantly reduced risk of acquiring LOS, both with CoNS and other bacteria. Perinatal inflammation may enhance the functional maturation of the preterm immune system and provide protection against LOS in high-risk preterm infants. *Pediatrics* 2012;129:e134–e141

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KEY WORDS

chorioamnionitis, late-onset sepsis, perinatal inflammation, premature infant

ABBREVIATIONS

BW—birth weight
CI—confidence interval
CLD—chronic lung disease
CoNS—coagulase-negative staphylococci
EOS—early-onset sepsis
GA—gestational age
HCA—histologic chorioamnionitis
HR—hazard ratio
LOS—late-onset sepsis
OR—odds ratio
TLR—Toll-like receptor

Dr Strunk performed the review of placental histology, data collection, and interpretation and wrote the article. Drs Burgner, Richmond, Simmer, and Charles helped design the study and contributed to data analysis, interpretation, and writing of the article. Dr Kohan was responsible for the management and retrieval of data from the neonatal database and contributed to the writing of the article. Dr Charles conducted the histologic analysis of placental specimens. Dr Doherty and Mrs Jacques performed the statistical analyses. Dr Burgner supervised all aspects of the study and wrote the article with Tobias Strunk. All authors approved the final version of the article before its submission.

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The incidence of premature delivery has increased by up to 30% over the past 2 decades in resource-rich countries. Premature infants account for 75% of neonatal morbidity and mortality, causing considerable human and health care costs, estimated at US\$26.2 billion per year in the United States alone.^{1–4}

Histologic chorioamnionitis (HCA, inflammation of the placental chorionic disk and the extraplacental membranes) is associated with preterm delivery and is inversely correlated with gestational age (GA) and birth weight (BW).⁵ Histologic evidence of chorioamnionitis is present in approximately 30% and 60% of placentae at 29 weeks and 23 to 24 weeks GA, respectively.^{5–9} Chorioamnionitis is associated with activation of fetal immune cells and increased levels of pro-inflammatory cytokines, including interleukins, tumor necrosis factor- α , and other mediators, in cord blood and cerebrospinal fluid.^{10–14} Perinatal inflammation can have deleterious effects, with increased neonatal morbidity, including intracranial hemorrhage, periventricular leukomalacia, and cerebral palsy.^{15–17} The infectious pathogens that drive perinatal inflammation also increase the risk of vertically acquired early-onset sepsis (EOS, with onset <72 hours of age).¹⁸ The presence of HCA has been associated with decreased risk of acute respiratory disease in preterm infants, at least partially due to stimulation of cortisol production and accelerated lung maturation, whereas the incidence of chronic lung disease (CLD) is increased by exposure to perinatal inflammation in most studies.^{9,18–26} Some studies suggest that HCA is more prevalent in survivors of extreme prematurity, and the improved outcome was largely attributable to decreased risk of respiratory disease.^{5,27}

Infection is one of the most common clinical problems encountered in the preterm population, who have greatly heightened susceptibility.^{28,29} In this

population, late-onset sepsis (LOS, age at onset >72 hours) is most frequently caused by coagulase-negative staphylococci (CoNS), and the incidence of infection is inversely correlated with GA and BW.³⁰ Infection with CoNS is associated with increased risk of common adverse neonatal outcomes such as CLD and cerebral palsy and results in prolonged hospital stay and health care costs.^{9,31–33}

Preterm birth is commonly complicated by perinatal inflammation of the maternal and/or fetal placental unit. Perinatal inflammation may have pervasive effects on immune function,³⁴ and we therefore hypothesized that chorioamnionitis may influence protective immune responses to CoNS and other pathogens causing LOS in preterm infants. Here we report an epidemiologic study of the relationship between HCA and the risk of LOS in a large cohort of very preterm infants.

METHODS

This single-center, retrospective cohort study was conducted at King Edward Memorial Hospital, the only tertiary perinatal center in Western Australia (total population ~2.1 million) with approximately 6000 deliveries annually. The study was approved by the institutional review board (#0830-02/08). All infants born at <30 weeks GA and admitted to the NICU at King Edward Memorial Hospital between January 1, 2001, and December 31, 2007, were identified from the neonatal database ($n = 1147$). Infants without placental histologic examination (usually those outborn and transferred postnatally) and those with major congenital anomalies that might increase susceptibility to infection were excluded.

Sepsis was defined by a combination of (1) suggestive clinical signs, (2) a positive blood culture (for the diagnosis of CoNS sepsis: only single-organism blood cultures were included),³⁰ and (3) appropriate antibiotic treatment of ≥ 5

days. Blood culture isolates were categorized into 2 groups: CoNS and other pathogenic bacteria. EOS was defined as sepsis with onset at <72 hours of age and LOS as sepsis with onset >72 hours of age. CLD was defined as a requirement for supplemental oxygen at 36 weeks postmenstrual age.

Placental Histology

Histopathologic scoring was performed on placentae that were collected and processed for histologic examination on all preterm infants born at <32 weeks GA as part of routine clinical practice. Semiquantitative histologic scoring of the extraplacental membranes, umbilical cord, chorionic plate, and placenta was performed using widely accepted guidelines³⁵ by an experienced perinatal histopathologist who was blinded to clinical outcomes. Evidence of a maternal inflammatory reaction was defined as an acute inflammatory reaction (predominantly neutrophils) in the decidua, extending into the chorion in the extraplacental membranes or from the maternal blood space into the chorionic plate. Fetal chorioamnionitis was defined as the presence of neutrophils marginating and infiltrating the umbilical vessels (usually the vein) into the cord or from the fetal vessels of the chorionic plate toward the amnion. Funisitis is inflammation involving the umbilical cord usually from the fetal vessels.

Preterm placentae are routinely cultured for microbial pathogens by taking a sample of the chorionic plate with overlying amnion. The microbiology laboratory culture the subamniotic layer for aerobic organisms including *Haemophilus*, mycoplasma, and ureaplasma species.³⁶

Statistical Analysis

Continuous data were summarized using means and standard deviations or medians and interquartile ranges, according to normality. Categorical data

were summarized using frequency distributions. Univariate group comparisons were performed using χ^2 and Fisher's exact tests as appropriate. Factors associated with EOS were evaluated using logistic regression analysis. Covariate effects were summarized using odds ratios (ORs) and their 95% confidence intervals (CIs). Time to first infection (CoNS or other LOS) and time until death were estimated using Kaplan-Meier survival probabilities. Effects of HCA and other covariates on the time until infection were examined using Cox proportional hazards regression, and their effects were summarized using hazard ratios (HRs) and 95% CIs. In the analyses of time to infection, deaths were censored. All evaluations of the effects of HCA were adjusted for GA. SPSS statistical software (version 15, SPSS, Chicago, IL) was used for data analysis. All hypothesis tests were two-sided, and P values $< .05$ were considered statistically significant.

RESULTS

Detailed placental histology was available from 869 of 1147 infants (75.7%) born during the study period. Of the 278 infants excluded because placental histologic examination was not performed, many were born in peripheral hospitals or by emergency delivery. An additional 31 infants were excluded for major congenital abnormalities that could significantly alter the risk of bacterial infection (eg, trisomy 21, gastroschisis, myelomeningocele). Therefore, 838 infants were included in the analysis. The mean GA of the study cohort was 26.9 ± 1.9 weeks with a mean BW of 936 ± 277 g. Ninety-six (11.5%) infants died, and of these 45 (46.9%) died on or before day 3 of life (Table 1).

HCA

On placental histologic examination, 384 of 838 (45.8%) had evidence of perinatal inflammation. Combined maternal and

TABLE 1 Basic Clinical Characteristics of Infants With Versus Without Exposure to Placental Inflammation

	Placental Inflammation		P Value
	No HCA $n = 535$ (64%)	HCA $n = 303$ (36%)	
GA	27.2 ± 1.6	26.3 ± 1.9	$<.001$
BW	950 ± 282	912 ± 267	.063
BW z score	-0.20 ± 1.04	0.09 ± 0.74	$<.001$
Gender (male)	293 (55)	154 (51)	.272
Apgar at 1 min <7	328 (62)	226 (75)	$<.001$
Apgar at 5 min <7	85 (16)	85 (28)	$<.001$
EOS	11 (2)	14 (5)	.036
Any LOS	180 (34)	96 (32)	.562
CoNS LOS	141 (26)	77 (25)	.765
CLD	182 (34)	110 (36)	.505
PVL	6 (1)	12 (4)	.006
Death	58 (11)	38 (13)	.458

Data shown as n (%) or mean $\pm$ SD. PVL, periventricular leukomalacia.

fetal HCA was observed in 132 (34.4%) placentae, maternal HCA alone in 154 of 384 (40.1%), and fetal HCA alone in 16 of 384 (4.2%). Funisitis in the absence of maternal HCA was uncommon (16 of 384, 4.2%). HCA was inversely related to GA; at 22 to 24 weeks GA, 86 of 152 (56.6%) had evidence of histologic HCA, compared with 32 of 142 (22.5%) at 29 weeks GA ($P < .0001$; Supplemental Fig 1).

Placental Culture and Histopathology

Placental microbiological cultures were performed in almost two-thirds of cases (536/838, 64%), and approximately one quarter of these (127, 23.7%) yielded a positive culture. *Ureaplasma urealyticum* was the most commonly isolated pathogen (53/127, 41.7%). HCA was significantly more common in cases of positive placental culture; maternal HCA was observed in 70.1% (89/127) of cases with positive placental cultures compared with 36.9% (151/409) of cases without bacterial growth ($P < .001$). Similarly fetal HCA or funisitis were more commonly associated with positive placental cultures (80/127, 63% with positive vs 135/409, 33% with negative cultures, $P < .001$).

Neonatal Sepsis

There were 328 episodes of neonatal sepsis in 300 (35.8%) infants. Three

percent (25/838) of infants developed EOS, with *Streptococcus agalactiae* and *Escherichia coli* the most common isolates. The basic characteristics of infants with and without EOS were similar (mean GA $26.4 [\pm 1.8]$ vs $26.9 [\pm 1.9]$ weeks, $P = .184$, and BW 889 g $[\pm 281$ g] vs 938 g $[\pm 277$ g], $P = .393$; Table 2). Infants with EOS had lower Apgar scores than those without (Apgar <7 at 1 minute in 88% vs 66%, $P = .029$, and Apgar <7 at 5 minutes in 42% vs 20%, $P = .009$, respectively; Table 2). Furthermore, infants with EOS had a higher incidence of any HCA (56% vs 36%, $P = .036$) and maternal HCA (56% vs 34%, $P = .020$) than infants without infection (Table 2). Importantly, none of the infants with EOS subsequently developed LOS.

Three hundred and four episodes of LOS occurred in 276 infants (32.9% of the total number of infants). Twenty-four of 276 (8.7%) infants developed 2 episodes of LOS, and 2 infants developed 3 episodes. The median age of onset of all LOS episodes was 10 days. For infants with >1 episode of LOS, only the first episode was included in the regression analysis. The majority of LOS cases were caused by CoNS (218 of 276 infants; 79%) with a median age at onset of 10 days. Sixty-eight of 276 cases (24.6%) of LOS were due to non-CoNS pathogens, of which *Staphylococcus aureus*, *E. coli*, *S. agalactiae*, and *Enterococcus* sp. were

the most frequent. The median age at onset of non-CoNS LOS was 13 days of age. Infants with LOS were born significantly earlier (mean GA 26.2 [$\pm$ 1.8] vs 27.2 [$\pm$ 1.8] weeks, $P < .001$) and with lower BW (848 g [$\pm$ 240 g] vs 980 g [$\pm$ 284 g], $P < .001$) than infants who did not develop LOS (Table 3). Infants who acquired LOS required respiratory support of longer duration (ventilated for a median of 297 hours vs 44 hours, $P < .001$, and continuous positive airway pressure for 736 hours vs 575 hours, $P = .001$). These infants also required parenteral nutrition of longer duration (16 days vs 10 days, $P < .001$), took longer to reach full enteral feeds (20 days vs 13 days, $P < .001$), and had a higher incidence of necrotizing enterocolitis (8% vs 3%, $P = .004$) and CLD (49% vs 28%, $P < .001$; Table 3).

Infants with CoNS LOS were more premature (26.2 [$\pm$ 1.8] vs 27.1 [$\pm$ 1.9] weeks, $P < .001$) and had lower BW than those without CoNS LOS (854g [$\pm$ 244 g] vs 965 g [$\pm$ 283 g], $P < .001$; Table 4). Furthermore, infants with CoNS LOS required respiratory support for longer (ventilated for 285 hours vs 49 hours, $P < .001$, and continuous positive airway pressure for 748 hours vs 584 hours, $P = .001$), needed parenteral nutrition for longer (16 days vs 10 days $P < .001$), and were slower to reach full enteral feeds (20 days vs 13 days, $P < .001$). In addition, infants with CoNS LOS had a higher incidence of necrotizing enterocolitis (7% vs 4%, $P = .029$) and CLD (48% vs 30%, $P < .001$; Table 4).

Chorioamnionitis and Risk of Neonatal Sepsis

The presence of any HCA was associated with increased risk of EOS (OR: 2.31 [95% CI: 1.03–5.15], $P = .036$). Similarly, maternal HCA was associated with an increased risk of EOS: more than half (14 of 25, 56%) of infants with EOS had evidence of maternal HCA compared

TABLE 2 Risk Factors for and Outcomes Associated With EOS

Risk Factor/Outcome	EOS		Unadjusted OR (95% CI) for EOS	P Value
	No, <i>n</i> = 813 (97%)	Yes, <i>N</i> = 25 (3%)		
GA	26.9 $\pm$ 1.9	26.4 $\pm$ 1.8	0.87 (0.71–1.07)	.184
BW	938 $\pm$ 277	889 $\pm$ 281	0.52 (0.12–2.32) ^a	.393
BW z score	−0.1 $\pm$ 0.9	−0.2 $\pm$ 1.12	1.09 (0.71–1.66)	.695
Gender (male)	435 (54)	12 (48)	0.80 (0.36–1.78)	.587
Apgar at 1 min <7	533 (66)	21 (88)	3.57 (1.06–12.08)	.029
Apgar at 5 min <7	160 (20)	10 (42)	2.87 (1.25–6.59)	.009
HCA	289 (36)	14 (56)	2.31 (1.03–5.15)	.036
Maternal chorioamnionitis	273 (34)	14 (56)	2.52 (1.13–5.62)	.020
Fetal chorioamnionitis	141 (17)	7 (28)	1.85 (0.76–4.51)	.135
Funinitis	189 (23)	10 (40)	2.18 (0.96–4.94)	.055
HCA + funinitis	175 (22)	8 (32)	2.18 (0.86–5.50)	.106
Death ^b	87 (11)	9 (36)	—	.001

Data shown as *n* (%) or mean $\pm$ SD.

^a OR for increase in kilograms.

^b Outcome variable in which EOS is a risk factor.

TABLE 3 Risk Factors for and Outcomes Associated With LOS

Risk Factor/Outcome	Any LOS		HR (95% CI)	P Value
	No, <i>n</i> = 562 (67%)	Yes, <i>n</i> = 276 (33%)		
GA	27.2 ($\pm$ 1.8)	26.2 ($\pm$ 1.8)	0.74 (0.69–0.79)	<.001
BW (g)	980 ($\pm$ 284)	848 ($\pm$ 240)	0.17 (0.10–0.27) ^a	<.001
BW z-score	−0.7 ($\pm$ 0.96)	−0.15 ($\pm$ 0.92)	0.93 (0.82–1.05)	.242
Gender (Male)	299 (53)	148 (54)	1.07 (0.84–1.36)	.909
Apgar at 1 min <7	368 (66)	186 (68)	1.22 (0.94–1.57)	.576
Apgar at 5 min <7	118 (21)	52 (19)	1.04 (0.77–1.41)	.436
HCA	207 (37)	96 (35)	0.97 (0.75–1.24)	.562
Maternal chorioamnionitis	197 (35)	90 (33)	0.96 (0.74–1.23)	.483
Fetal chorioamnionitis	103 (18)	45 (16)	0.92 (0.67–1.27)	.464
Funinitis	136 (24)	63 (23)	0.92 (0.70–1.22)	.701
HCA + Funinitis	125 (22)	58 (21)	0.91 (0.68–1.22)	.845
Ventilator	297 (53)	184 (67)	1.36 (1.06–1.74)	<.001
Ventilator (hours) ^b	44 (1–2415)	297 (1–1502)	1.02 (1.01–1.02) ^c	<.001
CPAP	308 (55)	179 (65)	1.07 (0.83–1.36)	.006
CPAP (hours) ^b	575 (1–1827)	736 (4–1856)	1.01 (1.00–1.01) ^c	.001
CLD	157 (28)	135 (49)	1.69 (1.33–2.14)	<.001
CVL	105 (19)	88 (32)	1.47 (1.14–1.89)	<.001
Total CVL days ^b	9 (1–39)	10 (1–38)	0.98 (0.95–1.01)	.454
Multiple CVL episodes	12 (11)	13 (15)	1.63 (0.93–2.85)	.491
Parenteral nutrition	335 (60)	189 (69)	1.13 (0.87–1.45)	.013
Total parenteral nutrition days ^b	10 (1–54)	16 (3–56)	1.04 (1.03–1.05)	<.001
Enteral (tube) feeds	304 (54)	178 (65)	1.06 (0.83–1.36)	.004
Time to full enteral feeds [hours] ^b	13 (4–59)	20 (6–133)	1.05 (1.04–1.06)	<.001
NEC ^d	18 (3)	21 (8)	—	.004
Death ^d	78 (14)	18 (8)	—	.002

Data shown as *n* (%), mean $\pm$ SD, or median (range). *P* values refer to the univariate analysis. CPAP, continuous positive airway pressure; CVL, central venous line; NEC, necrotizing enterocolitis.

^a HR for increase in kilograms.

^b Infants who did not require ventilation/CPAP/CVL/total parenteral feeding/enteral feeds were excluded.

^c HR for increase in days.

^d Outcome variables where LOS is a risk factor.

with 33.6% of infants without EOS (OR 2.52 [95% CI: 1.13–5.62], $P = .020$).

Chorioamnionitis significantly reduced the risk of LOS (Table 5 and Fig 1). LOS caused by any organism was less common in infants with evidence of perinatal

inflammation (adjusted HR for any HCA, HR: 0.74 [95% CI: 0.57–0.95], $P = .020$; for maternal HCA alone, HR: 0.73 [95% CI: 0.56–0.94], $P = .016$; for fetal HCA alone, HR: 0.67 [95% CI: 0.49–0.93], $P = .018$; Table 5).

TABLE 4 Risk Factors for and Outcomes Associated With CoNS LOS

Risk Factor/Outcome	CoNS LOS		HR (95% CI)	P Value
	No, n = 620 (74%)	Yes, n = 218 (26%)		
GA	27.1 (± 1.9)	26.2 (± 1.8)	0.74 (0.69–0.80)	<0.001
BW (g)	965 (± 283)	854 (± 244)	0.19 (0.11–0.32) ^a	<0.001
BW z score	−0.09 (± 0.96)	−0.12 (± 0.92)	0.96 (0.84–1.10)	0.710
Gender (male)	327 (53)	120 (55)	1.13 (0.87–1.48)	0.558
Apgar at 1 min <7	409 (67)	145 (67)	1.18 (0.89–1.57)	0.824
Apgar at 5 min <7	126 (21)	44 (20)	1.15 (0.83–1.60)	0.946
HCA	77 (35)	226 (37)	0.99 (0.75–1.31)	0.765
Maternal chorioamnionitis	216 (35)	71 (33)	0.96 (0.72–1.28)	0.544
Fetal chorioamnionitis	108 (17)	40 (18)	1.08 (0.76–1.52)	.764
Funisitis	147 (24)	52 (2)	0.99 (0.72–1.35)	.906
HCA + Funisitis	48 (22)	135 (22)	0.96 (0.70–1.34)	.883
Ventilator	335 (54)	146 (67)	1.38 (1.04–1.83)	.001
Ventilator (hours) ^b	49 (1–2415)	285 (1–1502)	1.02 (1.01–1.02) ^c	<.001
CPAP	344 (56)	143 (66)	1.11 (0.84–1.46)	.009
CPAP (hours) ^b	584 (1–1856)	748 (4–1601)	1.01 (1.00–1.02) ^c	.001
CLD	188 (30)	104 (48)	1.62 (1.24–2.12)	<.001
CVL	129 (21)	64 (29)	1.31 (0.98–1.75)	.010
Total CVL days ^b	10 (1–29)	10 (1–39)	0.97 (0.93–1.01)	.264
Multiple CVL episodes	15 (2)	10 (5)	1.62 (0.86–3.06)	.106
Parenteral nutrition	373 (60)	151 (69)	1.17 (0.88–1.56)	.017
Total parenteral nutrition (days) ^b	10 (1–56)	16 (3–56)	1.04 (1.03–1.05)	<.001
Enteral (tube) feeds	339 (55)	143 (66)	1.12 (0.85–1.48)	.005
Time to full enteral feeds (days) ^b	13 (4–59)	20 (6–133)	1.05 (1.04–1.06)	<.001
NEC ^d	23 (4)	16 (7)	—	.029
Death ^{cd}	81 (13)	15 (7)	—	.014

Data shown as n (%), mean ± SD, or median (range). P values refer to the univariate analysis. CPAP, continuous positive airway pressure; CVL, central venous line; NEC, necrotizing enterocolitis.

^a HR for increase in kilograms.

^b Infants who did not require ventilation/CPAP/CVL/total parenteral feeding/enteral feeds were excluded.

^c HR for increase in days.

^d Outcome variables where LOS is a risk factor.

TABLE 5 Histological Chorioamnionitis and Risk of LOS

Predictor Variables for Any LOS	N (%)	Adjusted for GA			Adjusted for GA and EOS		
		HR	95% CI	P Value	HR	95% CI	P Value
HCA	303 (36%)	0.74	0.57–0.95	.020	0.75	0.58–0.97	.028
Maternal chorioamnionitis	287 (34%)	0.73	0.56–0.94	.016	0.74	0.57–0.96	.024
Fetal chorioamnionitis	148 (18%)	0.67	0.49–0.93	.018	0.67	0.48–0.93	.016
Funisitis	199 (24%)	0.82	0.62–1.09	.173	0.83	0.63–1.10	.196
Predictor variables for CoNS LOS							
HCA	303 (36%)	0.74	0.57–0.95	.020	0.75	0.58–0.97	.028
Maternal chorioamnionitis	287 (34%)	0.73	0.55–0.98	.038	0.75	0.56–1.00	.051
Fetal chorioamnionitis	148 (18%)	0.82	0.58–1.17	.270	0.82	0.58–1.16	.258
Funisitis	199 (24%)	0.90	0.66–1.23	.490	0.91	0.66–1.24	.532
Predictor variables for other LOS							
HCA	303 (36%)	0.74	0.44–1.23	.243	0.75	0.45–1.26	.274
Maternal chorioamnionitis	287 (34%)	0.81	0.48–1.35	.417	0.83	0.49–1.38	.468
Fetal chorioamnionitis	148 (18%)	0.31	0.13–0.72	.006	0.31	0.13–0.72	.006
Maternal or fetal chorioamnionitis	303 (36%)	0.74	0.44–1.23	.243	0.75	0.45–1.26	.274
Funisitis	199 (24%)	0.75	0.42–1.33	.328	0.76	0.43–1.26	.353

Similarly, the risk of LOS with the most common pathogen, CoNS, was reduced in infants exposed to any HCA (HR: 0.74 [95% CI: 0.57–0.95], $P = .020$) as well as maternal HCA (HR: 0.73 [95% CI: 0.55–

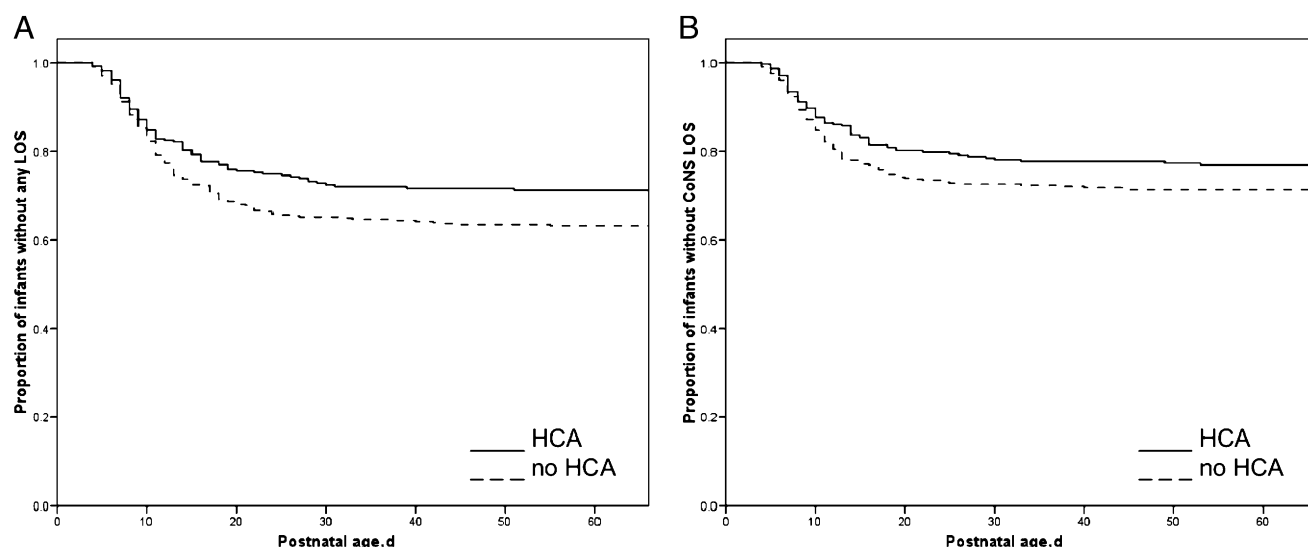
0.98], $P = .038$). The risk of LOS due to organisms other than CoNS was significantly reduced only by fetal chorioamnionitis (HR: 0.31 [95% CI: 0.13–0.72], $P = .006$; Table 5).

DISCUSSION

In this large single-center retrospective study, we report for the first time that HCA is associated with a significantly decreased risk of LOS in extremely preterm infants. Chorioamnionitis was associated with overall reduced risk of LOS and of infection with both the commonest cause of LOS in preterm infants, CoNS, and with infection due to other non-CoNS bacteria. The protective effect of HCA on LOS risk is opposite to the increased risk of EOS associated with HCA in the present and previous studies.^{18–20} Our data also confirmed the well-documented risk factors for LOS and CoNS infection: extreme prematurity and its complications and management.

Early-onset sepsis results from direct colonization and invasion of the neonate with the same organism that is recovered from the placenta, following vertical infection of the birth canal. The relatively low overall yield of routine bacterial cultures reflects the colonization and infection of the placenta with anaerobic and difficult-to-culture organisms, as suggested by recent data using 16sRNA polymerase chain reaction–based, culture-independent methods.³⁷ However, the pathogen most commonly cultured in placental inflammation related to preterm birth, *U. urealyticum*, does not typically cause acute clinical disease in the neonate. This suggests that if perinatal inflammation is due to a low virulence organism, it is relatively well controlled by the maternal immune system and does not result in bacterial invasion of the fetus and EOS, then it will enhance the maturation of the neonatal immune system. Uncontrolled infection with a virulent organism will result in direct infection of the fetus, clinically presenting as EOS.

Conversely the organisms causing perinatal inflammation rarely result in clinical LOS. Instead it appears that chorioamnionitis has significant effects on immune function and therefore

**FIGURE 1**

Effect of HCA on the incidence of LOS. Kaplan-Meier survival curves indicate the proportion of infants without LOS versus postnatal age (days). A, LOS with any organism; B, LOS with CoNS.

indirectly modulates the risk of LOS with nosocomial organisms. The mechanisms through which HCA may result in maturation of the fetal and neonatal immune system are unknown but may involve transfer of bacterial ligands and/or maternal immune mediators across the placenta. Recent animal data demonstrate that exposure to specific *Toll*-like receptor (TLR) ligands before induced polymicrobial sepsis significantly improves bacterial control and survival of neonatal mice.³⁸ Intra-amniotic injection of lipopolysaccharide in sheep produces chorioamnionitis and amplifies monocyte responsiveness to TLR agonists.³⁹ Therefore, specific TLR agonists may be useful therapeutic tools to accelerate the maturation of neonatal immunity, once the underlying pathways and the potential risks are fully understood.

Activation of the fetal immune system by perinatal inflammation is increasingly recognized, and the effects of early life influences such as inflammation on the developing immune phenotype may be long lasting.⁴⁰ High levels of cord blood pro-inflammatory cytokine levels are associated with a decreased risk of allergic disease in later childhood,⁴¹ and offspring of allergic mothers who

developed neonatal sepsis after term delivery had a reduced incidence of allergic diseases compared with those without neonatal infection.³⁴ Other studies suggest that the presence of chorioamnionitis in preterm infants may be associated with increased risk of recurrent wheezing or asthma in childhood.^{42,43} In summary, these data suggest that perinatal inflammation has significant, GA- and context-dependent effects on the maturation of the immune system and clinical outcomes.

Despite the reduced risk of LOS and CLD in infants exposed to perinatal inflammation, there are also adverse associations with chorioamnionitis. For example, elevated cytokine levels in cord blood are associated with significantly increased risk of developing cerebral palsy.^{10,11,13,14,44} Moreover, we have previously described dysfunctional immune responses and aberrant regulation of inflammation in term and preterm infants.^{45,46} The complex effects of perinatal inflammation, which complicates the majority of extreme preterm deliveries, on both postnatal immune function and clinically relevant outcomes clearly warrant additional investigation

We acknowledge that our study has some limitations, including the availability of placental histopathology in 75% of all babies, which could theoretically bias the findings. The sample size of our study did not allow us to analyze the effect of HCA on long-term neurologic outcomes, such as intracranial hemorrhage or periventricular leukomalacia. The strengths of our study include the relatively large sample size for a single-center study, yielding clear results. This retrospective cohort from the postsurfactant era was almost universally exposed to antenatal steroids, standard management for threatened preterm delivery. The epidemiologic characteristics of our cohort and the incidence of HCA are similar to those previously reported,^{5,9} and our findings are likely to be broadly relevant. Confirmation in an independent cohort is required.

In conclusion, this single-center, retrospective cohort study demonstrates that HCA is associated with significantly reduced risk of LOS in high-risk preterm infants. Understanding the mechanisms through which perinatal inflammation affects postnatal immune function is important in improving the management of

infection and other adverse outcomes in extremely preterm infants.

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